

End-to-End House Price Prediction using Linear Regression

1. Problem Definition

The goal of this project is to build a supervised machine learning model that predicts housing prices based on structural and neighborhood characteristics from the Boston Housing dataset. The dataset contains 506 observations with 13 numerical features describing housing conditions such as crime rate, number of rooms, pollution level, and socioeconomic indicators. The target variable is MEDV, which represents the median value of owner-occupied homes.

This problem is formulated as a regression task because the objective is to predict a continuous numerical value (house price).

2. Approach

The project follows a complete machine learning pipeline:

- **Data Exploration:** Summary statistics, feature ranges, and correlation analysis were conducted to understand relationships between variables and house prices.
 - **Feature Engineering:** Input features were standardized to improve the performance and stability of gradient descent optimization.
 - **Model Implementation:** A linear regression model was implemented from scratch using vectorized gradient descent. The dataset was split into training and testing sets to evaluate model performance.
 - **Evaluation:** Model performance was evaluated using Mean Squared Error (MSE) and the coefficient of determination (R^2).
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3. Results

The trained model achieved an R^2 score of **0.65**, indicating that approximately **65% of the variance in housing prices** can be explained by the features in the dataset.

The analysis showed that several features significantly influence housing prices:

- RM (number of rooms) positively affects house prices
- LSTAT (percentage of lower-income population) negatively affects prices
- Environmental and neighborhood variables such as pollution (NOX) and crime rate (CRIM) also influence property value.

These findings align with expected economic patterns in real estate markets.

4. Bias–Variance Discussion

Linear regression is a relatively simple model and may exhibit higher bias because it assumes a linear relationship between features and the target variable. While the model captures major trends in the data, it may fail to represent complex nonlinear relationships.

Regularization techniques such as L2 regularization help reduce overfitting by penalizing large coefficients, improving the model's generalization ability. However, excessive regularization may increase bias by oversimplifying the model.

Overall, the results demonstrate a balance between bias and variance suitable for a baseline predictive model.